

Final Report

What Makes a Developer Earn More?

The Role of Language Popularity and Structural Factors in Salary Formation

Course	DSA210 Introduction to Data Science
Term	Spring 2025-2026
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Repository	https://github.com/NilKadakal/DSA210_NilKadakal
Interactive Website	https://nilkadakal.github.io/DSA210_NilKadakal/website/
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This report presents the complete data science pipeline of the project: motivation, data sources, data preparation, enrichment, exploratory analysis, hypothesis testing, machine learning modeling, findings, limitations, and future work.

Executive Summary

This project investigates the factors associated with higher developer salaries using the Stack Overflow Developer Survey 2024 enriched with programming-language popularity scores from the TIOBE Index. The central research question is whether programming language choice and market popularity influence developer compensation after accounting for structural variables such as professional experience, education, country, and work modality.

The results show that programming language choice and work modality are statistically associated with salary differences. However, TIOBE popularity is not a strong standalone predictor of salary. The Spearman correlation between TIOBE popularity and median salary by language is not statistically significant. Machine learning results further support this conclusion: adding language and TIOBE features slightly improves Ridge Regression performance, but the improvement is small. The strongest predictive factors are country and experience, especially country-level labor-market context and years of coding experience.

The final interpretation is that developers should not treat language popularity as a direct salary signal. Salary formation appears to depend more strongly on structural and contextual factors, including geography, seniority, work conditions, and specialization.

1. Motivation

The main motivation behind this project is to understand which factors are associated with higher developer salaries in the software industry. As a Computer Science and Engineering student, I am personally interested in how technical skills, programming language choices, work conditions, and experience levels shape career outcomes. In technology-related career discussions, programming languages are often treated as direct indicators of market value.

For example, it is common to hear claims such as "learning a popular language leads to a better salary" or "some languages are automatically more valuable than others." However, salary formation is likely more complex than language choice alone. This project therefore investigates whether programming language popularity has a meaningful relationship with developer compensation after considering broader labor-market and professional factors.

Research question: To what extent do programming language choice and market popularity influence developer salaries after controlling for structural variables such as experience, education, country, and work modality?

Hypothesis: Programming language choice, years of professional experience, education level, and work modality significantly explain salary differences; however, programming language popularity, measured through the TIOBE Index, has a positive but limited marginal effect once structural factors are considered.

2. Data Source and Collection

The main dataset used in this project is the Stack Overflow Developer Survey 2024. This dataset contains developer-level responses on salary, programming languages, years of coding experience, professional experience, country, education level, employment status, developer role, work modality, and organization size. It is suitable for this project because it combines technical skill variables with structural career-related variables.

Since the project uses a publicly available dataset, it is enriched with an additional external dataset: the TIOBE Index 2024. The TIOBE Index was used as a proxy for programming language popularity. Monthly TIOBE ratings were collected and transformed into yearly average popularity scores for programming languages. These yearly popularity values were then merged with Stack Overflow survey records after language-name normalization.

The full raw and processed files are large, so the repository includes sampled processed datasets and points to external storage for the complete datasets. This preserves reproducibility while keeping the GitHub repository lightweight.

2.1 Dataset Summary After Cleaning

Metric	Annual Salary	Log Salary	Professional Experience
count	21,025	21,025	21,025
mean	\$78,358	10.84	9.85
std	\$61,435	1.15	8.29

Metric	Annual Salary	Log Salary	Professional Experience
min	\$261	5.56	0.50
25%	\$34,370	10.44	4.00
50%	\$65,000	11.08	7.00
75%	\$107,406	11.58	14.00
max	\$382,164	12.85	50.00

Table 1. Summary statistics for the cleaned salary dataset used in EDA and modeling.

3. Data Preparation and Enrichment

The data preparation process was designed to make the survey records suitable for statistical testing and machine learning. The raw survey includes missing salaries, textual experience values, multi-label programming-language fields, and extreme compensation values. These issues were handled before the analysis.

- Loaded the Stack Overflow Developer Survey 2024 dataset and selected variables related to salary, experience, country, education, work modality, developer role, organization size, and programming languages.
- Filtered for employed respondents with valid positive annual compensation values.
- Removed extreme salary observations using a 1st to 99th percentile filter to reduce distortion from unrealistic or highly unusual values.
- Converted professional experience values such as "Less than 1 year" and "More than 50 years" into numeric approximations.
- Applied a natural-log transformation to annual salary because salary values are strongly right-skewed.
- Split multi-language responses into language-level rows so that each developer-language pair could be connected to a TIOBE popularity score.
- Normalized language names between Stack Overflow and TIOBE and merged yearly average TIOBE ratings into the language-level dataset.
- Saved output tables and figures under the repository structure for reporting and presentation.

4. Exploratory Data Analysis

The exploratory analysis focused on understanding salary distributions and identifying major relationships before formal hypothesis testing and machine learning. The main EDA dimensions were salary distribution, professional experience, work modality, programming language, and the relationship between language popularity and median salary.

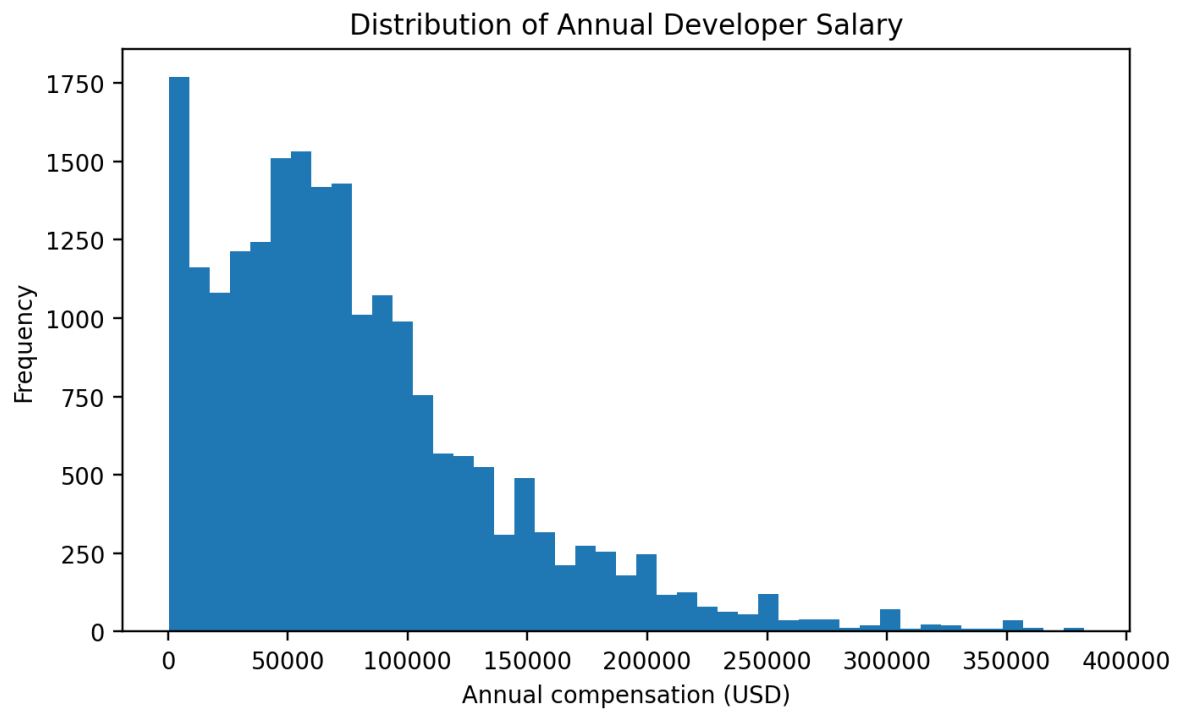


Figure 1. Annual salary distribution after cleaning and outlier filtering. The distribution remains right-skewed, supporting the use of log salary in later analysis.



Figure 2. Salary generally rises with professional coding experience, but variation remains substantial within similar experience levels.

4.1 Work Modality

The work-modality comparison shows a clear salary difference between remote, hybrid, and in-person respondents. Remote developers have the highest median compensation in the cleaned dataset, followed by hybrid and then in-person developers. This pattern supports testing whether work modality is associated with salary differences, while also requiring careful interpretation because remote work can be correlated with country, seniority, job type, and access to international labor markets.

Work Modality	Count	Median Salary	Mean Salary
Remote	8,043	\$75,184	\$89,358

Work Modality	Count	Median Salary	Mean Salary
Hybrid (some remote, some in-person)	9,261	\$66,228	\$77,371
In-person	3,721	\$44,681	\$57,036

Table 2. Salary by work modality.

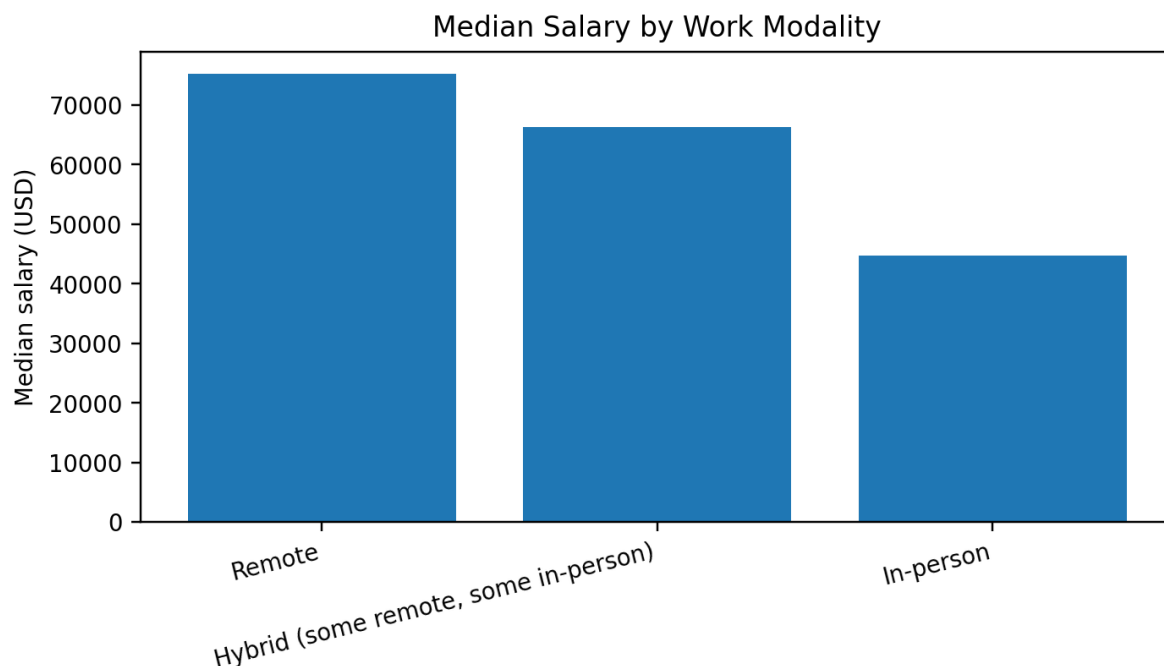


Figure 3. Median salary by work modality.

4.2 Programming Languages and Salary

The language-based analysis shows that salaries differ across programming languages. Some languages are associated with higher median salaries even when they are not among the most popular languages. Conversely, widely used languages are not automatically the highest-paying languages. This distinction is central to the project because it separates language popularity from salary value.

Language	Count	Median Salary	Average TIOBE Rating
JavaScript	13,216	\$63,694	3.502
SQL	11,571	\$64,691	1.813
HTML/CSS	10,881	\$61,770	
Python	10,373	\$67,666	17.984
TypeScript	8,979	\$65,799	
Bash	7,764	\$75,184	
Java	6,062	\$60,441	9.063
C#	6,002	\$65,514	6.400
C++	3,953	\$63,694	10.392
PHP	3,601	\$50,000	1.337

Table 3. Ten most common languages in the language-level dataset.

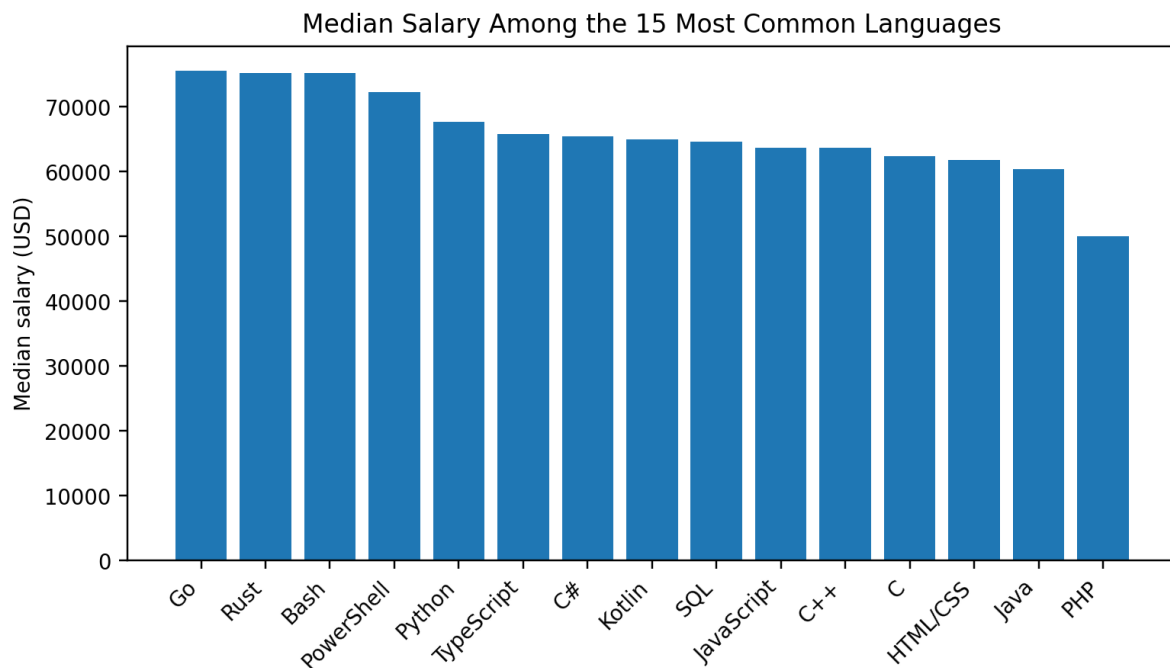


Figure 4. Median salary among the 15 most common programming languages.

4.3 Language Popularity vs Salary

TIOBE popularity does not show a clear monotonic relationship with median salary. For example, Python has a high TIOBE rating, but other languages with much lower ratings can have equal or higher median salaries. This suggests that popularity and compensation are different concepts: a language may be widely used because it is general-purpose, easy to learn, or common in entry-level contexts, while higher salaries may be tied to specialization, role seniority, industry, or geography.

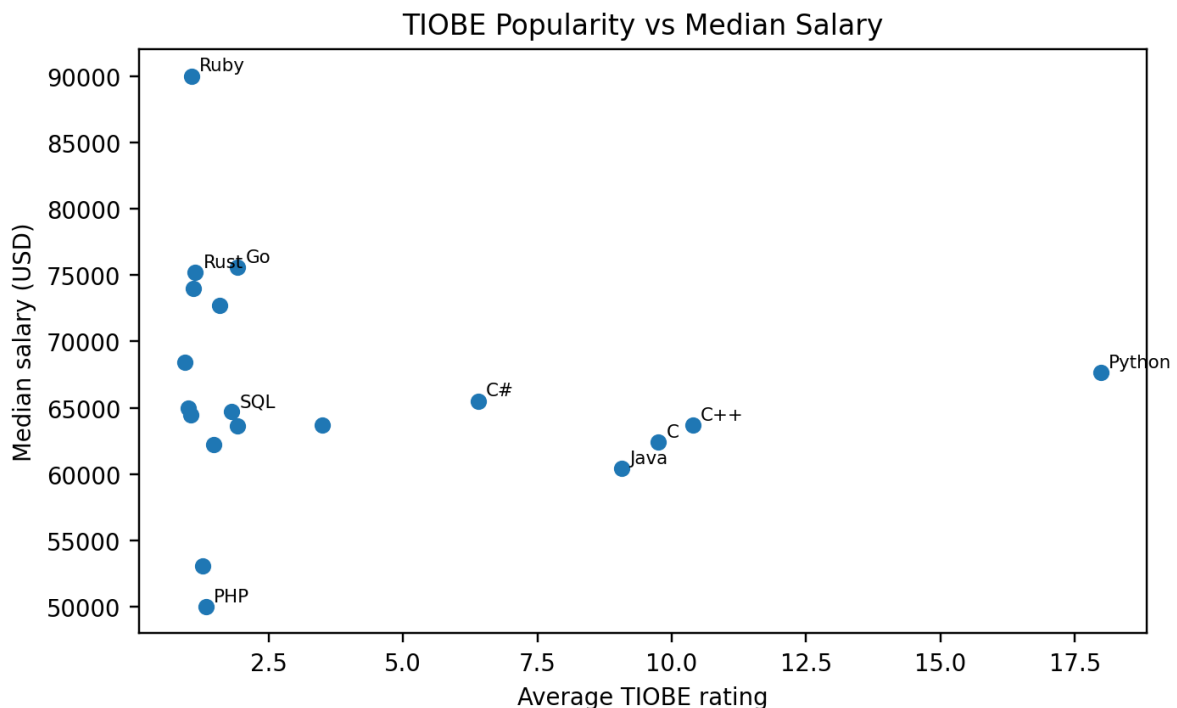


Figure 5. TIOBE popularity versus median salary by programming language.

5. Hypothesis Testing

Formal statistical tests were used to evaluate whether the relationships observed during EDA were statistically meaningful. The project used one-way ANOVA for language groups, Welch two-sample t-test for remote versus in-person

work, and Spearman rank correlation for TIOBE popularity versus median salary.

Test	Statistic	p-value	Interpretation
ANOVA LogSalary Across Languages	38.134	8.75e-321	Reject H0. Mean log salaries differ significantly across language groups.
Welch TTest LogSalary Remote vs InPerson	22.093	1.16e-104	Reject H0. Remote and in-person log salaries differ significantly.
Spearman TIOBE vs MedianSalary	-0.272	2.60e-01	Fail to reject H0. No significant monotonic relationship is found.

Table 4. Hypothesis test results.

The ANOVA result indicates that programming language choice is associated with salary differences. The Welch t-test indicates that work modality is also associated with salary differences. However, the Spearman test does not provide evidence that TIOBE popularity has a significant monotonic relationship with median salary across languages. This is one of the central findings of the project.

6. Machine Learning Analysis

The machine learning component extends the statistical analysis by comparing baseline models with extended models that incorporate programming language and TIOBE popularity features. The target variable is log-transformed yearly compensation. This target is more stable than raw salary because compensation values are highly skewed.

Two model families were used: Ridge Regression and Random Forest Regressor. The baseline models used structural and demographic variables such as country, professional coding experience, total coding experience, age group, remote work status, education level, developer type, and organization size. The extended models added programming-language information, average TIOBE rating, and TIOBE observation count.

Because the dataset was expanded to language-level rows, the same respondent could appear in multiple rows. To avoid data leakage, the train-test split was performed using Responseld as a grouping variable, ensuring that the same developer did not appear in both training and testing sets.

Model	MAE log	RMSE log	R2 log	MAE USD	RMSE USD
Extended Ridge Regression + TIOBE	0.554	0.875	0.481	\$29,946	\$46,477
Baseline Ridge Regression	0.555	0.879	0.476	\$29,989	\$46,420
Baseline Random Forest	0.565	0.883	0.472	\$30,235	\$45,899
Extended Random Forest + TIOBE	0.566	0.884	0.470	\$30,207	\$46,080

Table 5. Machine learning model comparison.

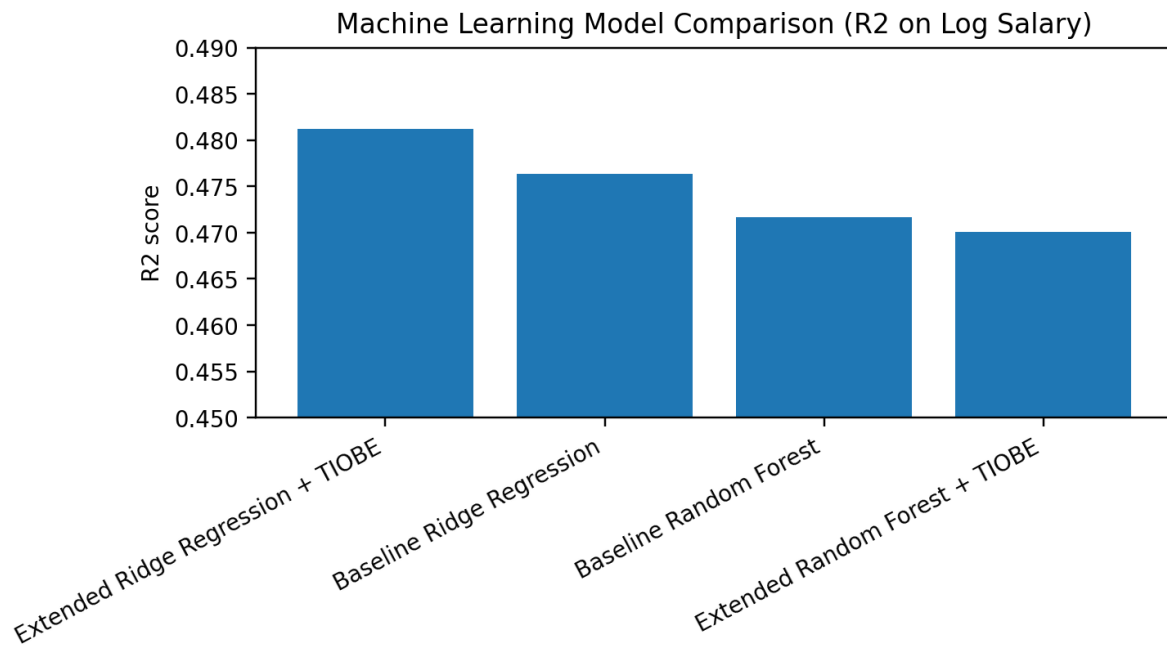


Figure 6. R2 comparison across baseline and extended machine learning models.

The best-performing model is Extended Ridge Regression + TIOBE, with an R2 score of 0.481 on log salary. The baseline Ridge model reaches 0.476, so adding language and TIOBE features improves predictive performance slightly. However, the gain is small, indicating that language popularity has limited marginal predictive value after structural variables are already included.

6.1 Feature Importance

Random Forest feature importance provides an interpretable view of which variables contribute most strongly to prediction. The strongest predictors are country and experience-related variables. Average TIOBE rating appears among the top features, but its importance is much lower than country and experience.

Feature	Importance
Country: United States	0.268
Total coding experience	0.171
Professional coding experience	0.166
Country: other/infrequent	0.062
Country: Ukraine	0.047
Country: India	0.044
Country: Brazil	0.028
Average TIOBE rating	0.023
Age: 18-24	0.012
Remote work: In-person	0.012

Table 6. Top Random Forest feature importances.

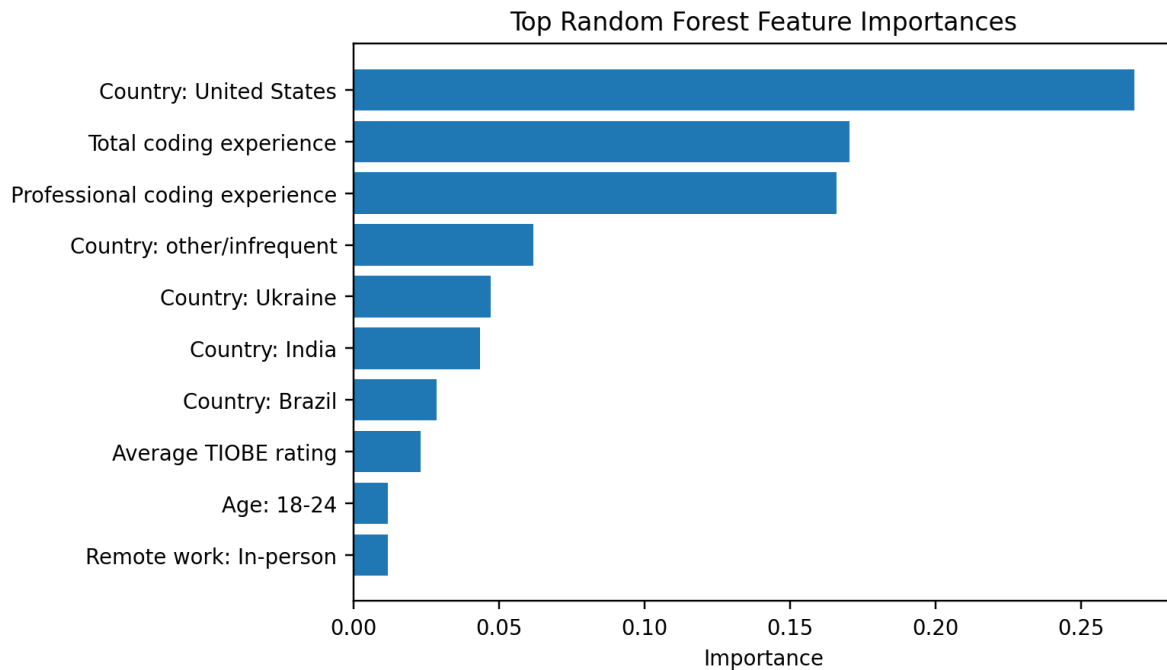


Figure 7. Country and experience dominate the Random Forest feature importance ranking.

7. Interactive Project Website

As additional presentation material, an interactive project website was created and deployed through GitHub Pages. The website summarizes the core project conclusion, the best-performing machine learning model, model comparison results, Random Forest feature importance, key findings from EDA and hypothesis testing, and a scenario-based salary estimator.

The estimator should not be interpreted as an exact salary prediction system. It is a transparent what-if interface calibrated to the project findings. Its purpose is to communicate the main conclusion that country and experience are stronger salary predictors than programming language popularity alone.

Website: https://nilkadakal.github.io/DSA210_NilKadakal/website/

8. Findings

- **Programming language choice is associated with salary differences.** The ANOVA test shows statistically significant salary differences across language groups.
- **Work modality is associated with salary.** The Welch t-test shows a significant difference between remote and in-person developers, although this should not be interpreted as direct causation.
- **Programming language popularity is not a strong standalone salary predictor.** The Spearman correlation between TIOBE popularity and median salary is not statistically significant.
- **Adding TIOBE features provides only a small ML improvement.** Extended Ridge Regression improves R2 from 0.476 to 0.481.
- **Structural factors dominate salary prediction.** Country and experience are much stronger predictors than TIOBE popularity in the feature importance results.
- **The original hypothesis is partially supported.** Language choice, experience, and work modality matter, while language popularity has only a limited marginal effect.

9. Personal Observations

One of my main observations during this project was that salary prediction is more complex than it appears at first. At the beginning, I expected programming language popularity to have a clearer relationship with compensation. However, the

analysis showed that popularity does not directly translate into higher salary. A language can be popular because it is widely used in general-purpose, educational, or entry-level contexts. This does not automatically make it highly paid.

Another important observation was the strength of country-level effects. The feature importance results showed that geography is one of the most powerful predictors of salary. This made it clear that compensation is not only about individual technical skills; it is also shaped by the economic and labor-market environment in which a developer works.

I also observed that model complexity does not guarantee better performance. Random Forest is more flexible than Ridge Regression, but it did not perform better in this project. This helped me understand that simpler models can sometimes be more effective when the strongest signals in the data are broad structural factors.

10. Limitations and Future Work

This project has several limitations that should be considered when interpreting the results.

- **Self-reported data:** Salary, experience, job role, and other variables may contain reporting errors or inconsistencies.
- **Cross-sectional design:** The dataset captures respondents at one point in time, so the analysis cannot establish causal relationships.
- **Country and cost-of-living effects:** Country is highly predictive, but the project does not fully adjust for taxation, cost of living, purchasing power, or local labor-market conditions.
- **Single popularity metric:** TIOBE is useful as an external enrichment source, but it may not fully represent labor-market demand, job-posting volume, GitHub activity, or salary premiums.
- **Multi-language complexity:** Developers often use several languages, and salary may depend more on domain specialization, seniority, industry, or company characteristics than on any single language.

Future work could extend the project by adding purchasing-power or cost-of-living adjustments, comparing multiple years of Stack Overflow survey data, incorporating job-posting datasets, testing models such as Gradient Boosting or XGBoost, building role-specific models, and exploring interaction effects between country, experience, remote work, and language choice.

11. Conclusion

This project investigated whether programming language popularity and structural factors explain developer salaries. The results show that programming language choice and work modality are associated with salary differences, but language popularity alone is not a strong predictor of compensation. The machine learning results support this conclusion: adding TIOBE popularity features slightly improves Ridge Regression, but the improvement is small.

The strongest predictors are country and experience, indicating that salary formation is shaped primarily by structural labor-market factors. The final conclusion is that developers should not interpret programming language popularity as a direct salary signal. Popularity may reflect broad usage, but salary formation depends more heavily on geography, experience, work conditions, specialization, and broader market context.

12. Project Materials and Reproducibility

The project repository contains the Python code, notebooks, generated output tables, generated figures, README documentation, proposal document, final report, and website source file.

Material	Location
GitHub Repository	https://github.com/NilKadaka/DSA210_NilKadaka
Interactive Website	https://nilkadaka.github.io/DSA210_NilKadaka/website/
Main EDA and Hypothesis Testing Notebook	Developer_Salary_Analysis.ipynb
Machine Learning Notebook	ml_salary_models.ipynb
Python Scripts	src/prepare_tiobe.py and src/salary_analysis.py
Output Tables	outputs/tables/
Generated Figures	outputs/figures/

Material	Location
Website Source	website/index.html
Dependencies	requirements.txt

Table 7. Main project materials.

To reproduce the analysis, the raw Stack Overflow survey file should be placed under `data/raw/survey_results_public.csv`. TIOBE monthly files should be placed under `data/raw/` using the expected file pattern, then `src/prepare_tiobe.py` should be run before `src/salary_analysis.py`. Machine learning results are produced in `ml_salary_models.ipynb`.

13. Academic Integrity and AI Assistance Disclosure

This project is original work prepared for DSA210 Introduction to Data Science. The analysis decisions, code execution, statistical interpretation, machine learning evaluation, repository organization, and final project materials were reviewed and completed by the author.

AI tools were used only as support tools for writing assistance, debugging support, code-structure suggestions, explanation refinement, README/report organization, and presentation polishing. Representative prompt categories included requests to structure the README, explain hypothesis-test results in academic English, debug preprocessing and modeling issues, improve limitations/future-work wording, and check guideline compliance. AI-generated suggestions were edited, validated, and integrated by the author rather than accepted blindly.

14. References and Data Sources

- Stack Overflow Developer Survey 2024 public dataset: <https://survey.stackoverflow.co/2024/>
- TIOBE Index: <https://www.tiobe.com/tiobe-index/>
- TIOBE index ratings repository used for language popularity enrichment: <https://github.com/toUpperCase78/tiobe-index-ratings>
- Project repository: https://github.com/NilKadikal/DSA210_NilKadikal
- Python libraries used: pandas, NumPy, matplotlib, SciPy, scikit-learn, Jupyter.